Generative summary

Computer Organization and Architecture

(Code: CS256)

Lecture-26: Internal Memory Design

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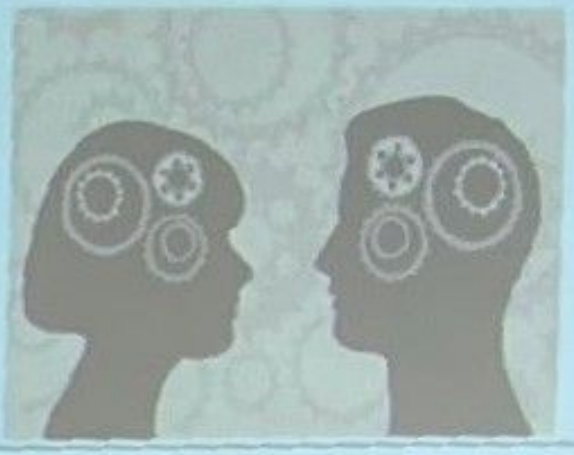
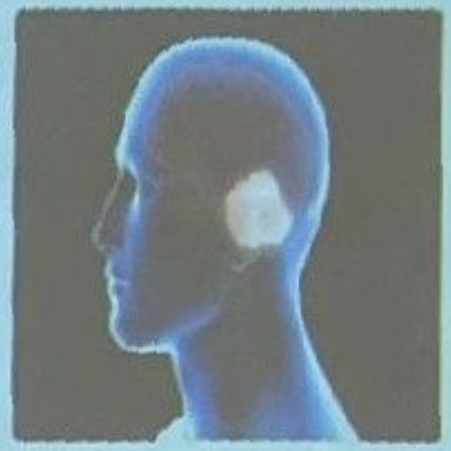
Memory

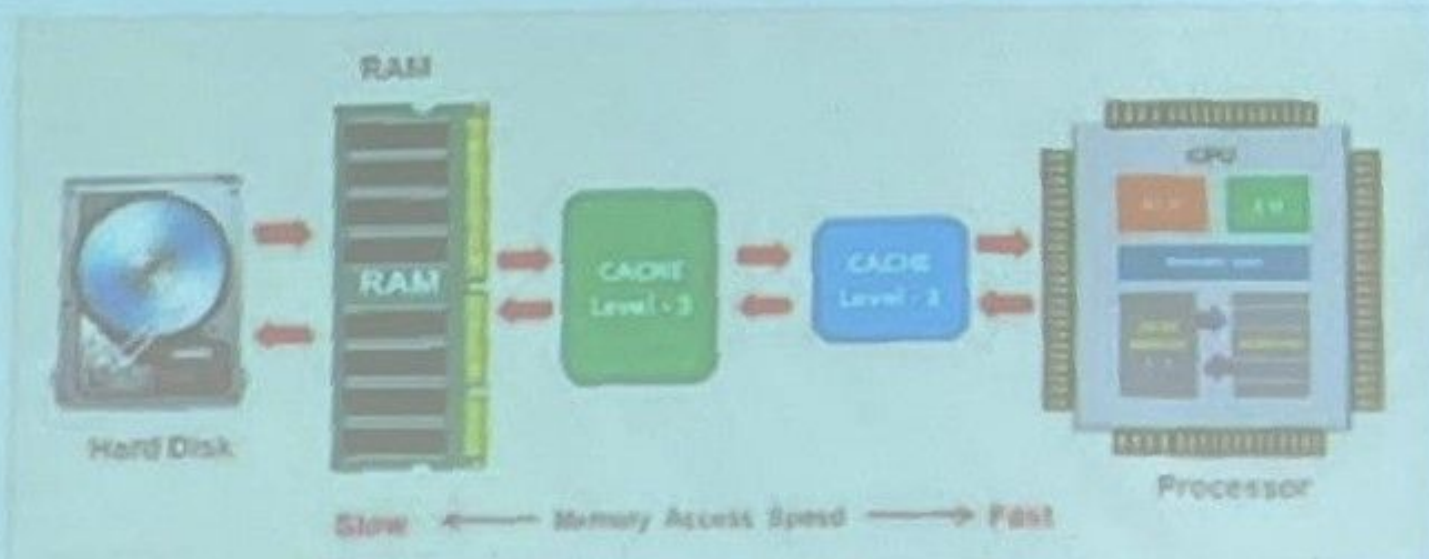
- *Computer memory is the storage space in the computer, where data and instructions required for processing are stored.*
- The memory is divided into large number of small parts called blocks or locations.
- Each location or block has a unique address, which varies from zero to memory size minus one.
- Classifications: Internal and External Memory



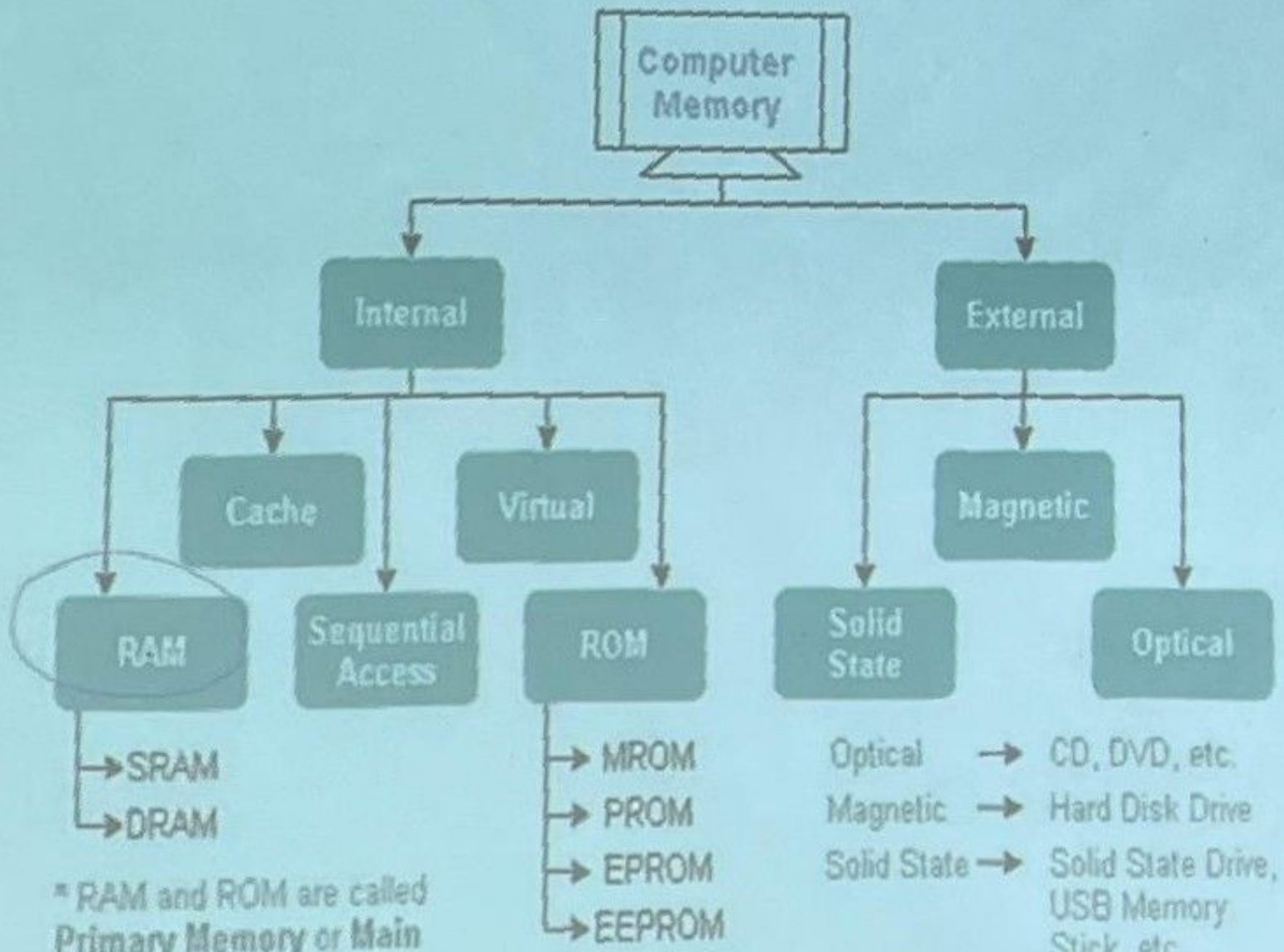
Location



- CPU
  - Registers, Control Memory
- Internal
  - Cache
  - Main Memory – RAM
  - ROM
- External
  - Disk, Tape, …



Memory Classification



* RAM and ROM are called **Primary Memory** or **Main Memory**.

Memory Technology

- Transistor Logic
  - Dynamic Memory Technology
  - E.g RAM

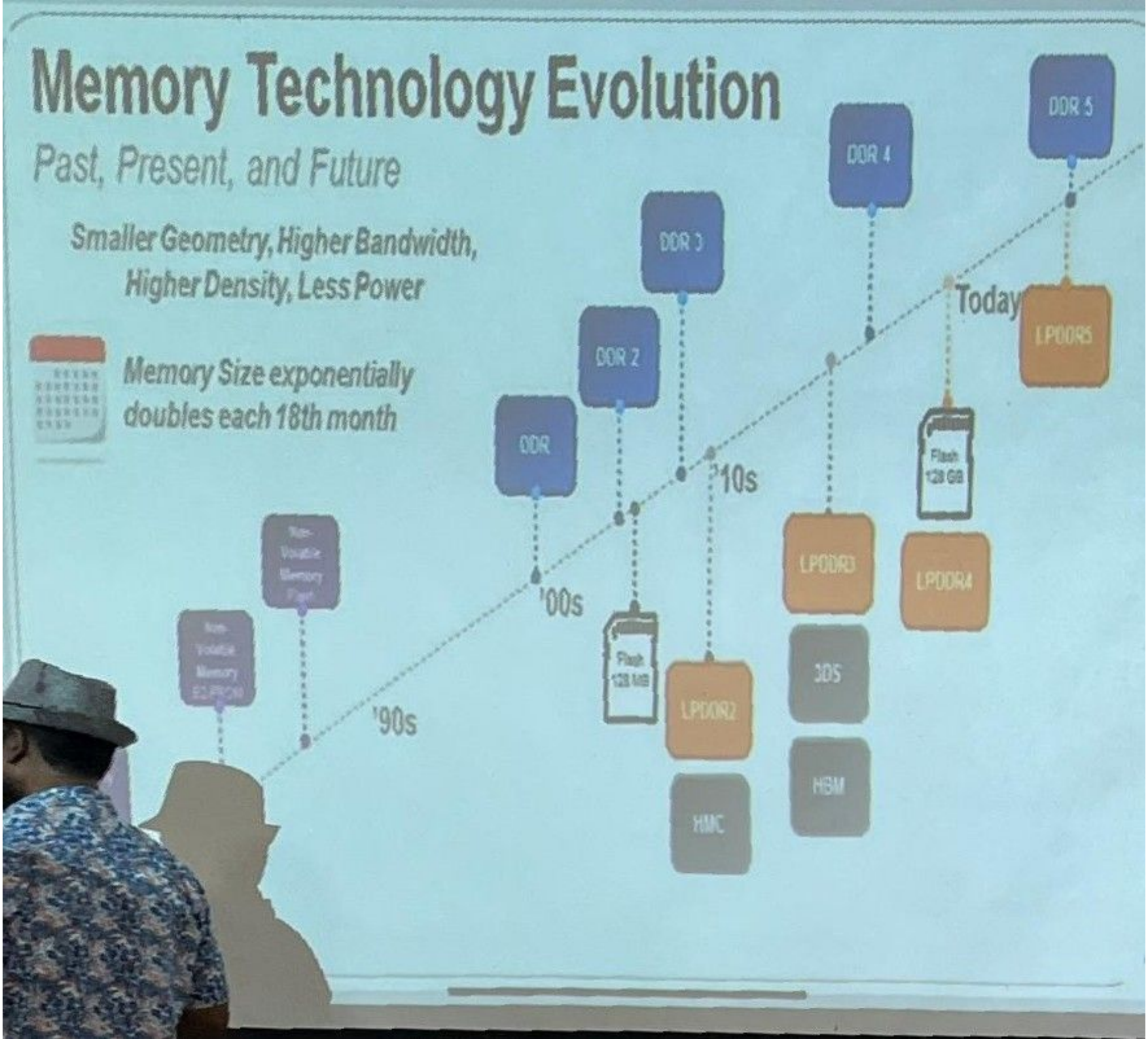
- Semiconductor Logic.
  - Static Memory Technology
  - E.g Cache

Memory Technology Evolution

Past, Present, and Future

Smaller Geometry, Higher Bandwidth, Higher Density, Less Power

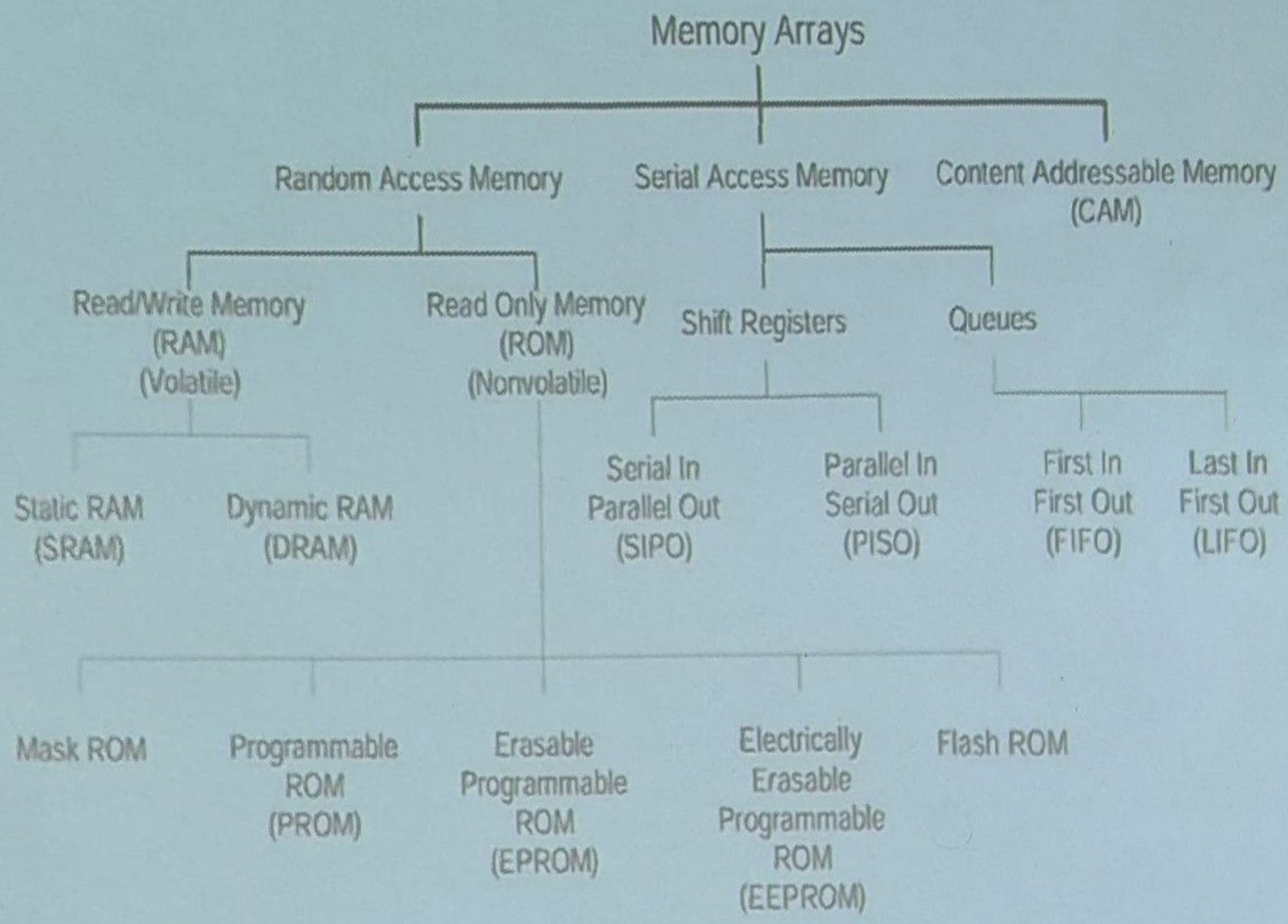
Memory Size exponentially doubles each 18th month



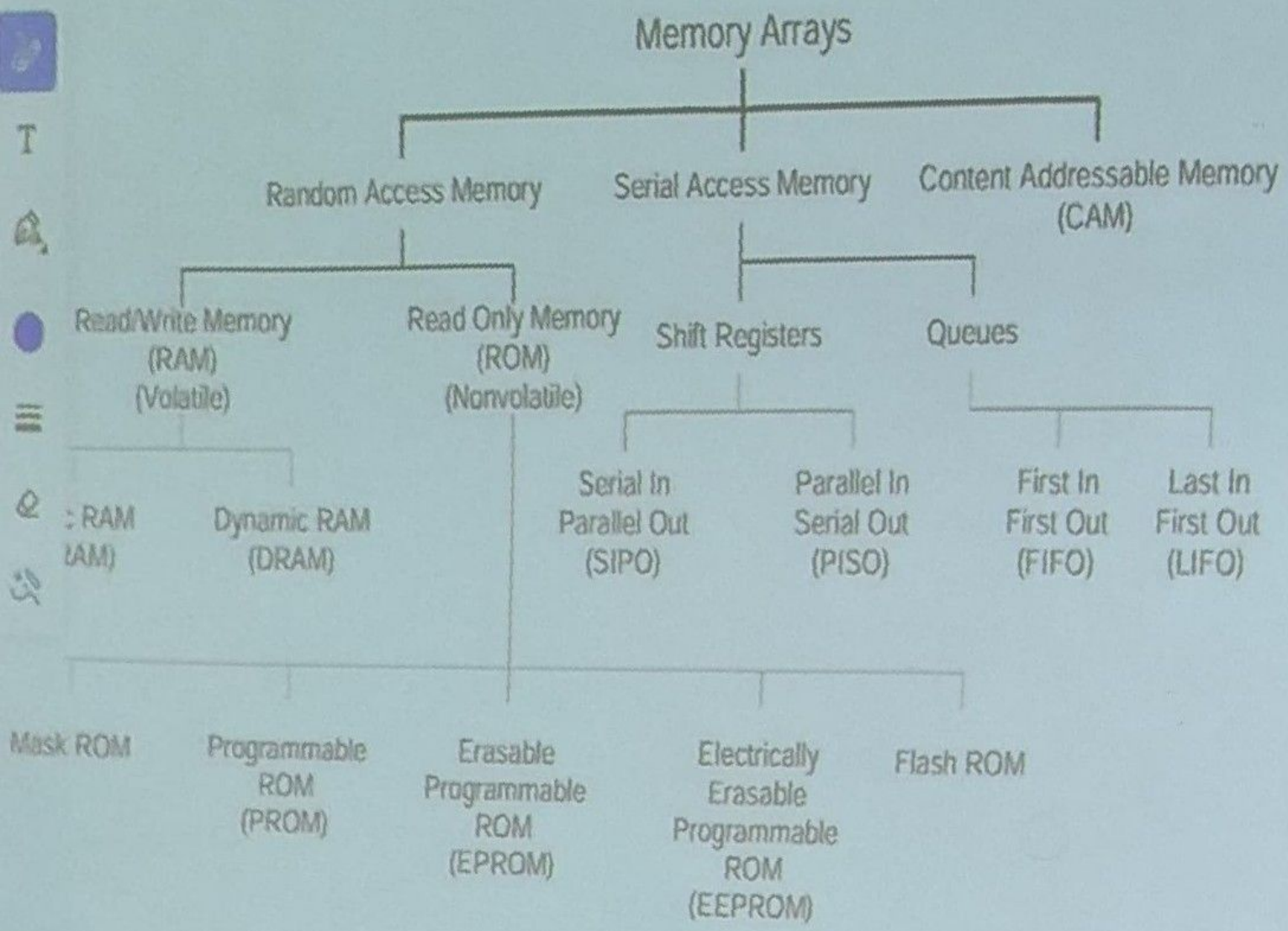
Key Aspects of Semiconductor Technology

- Organization
- RAM
- ROM
- Chip Logic
- Chip Packaging
- Module Organization
- Interleaved Memory

Internal Memory Arrays



Internal Memory Arrays



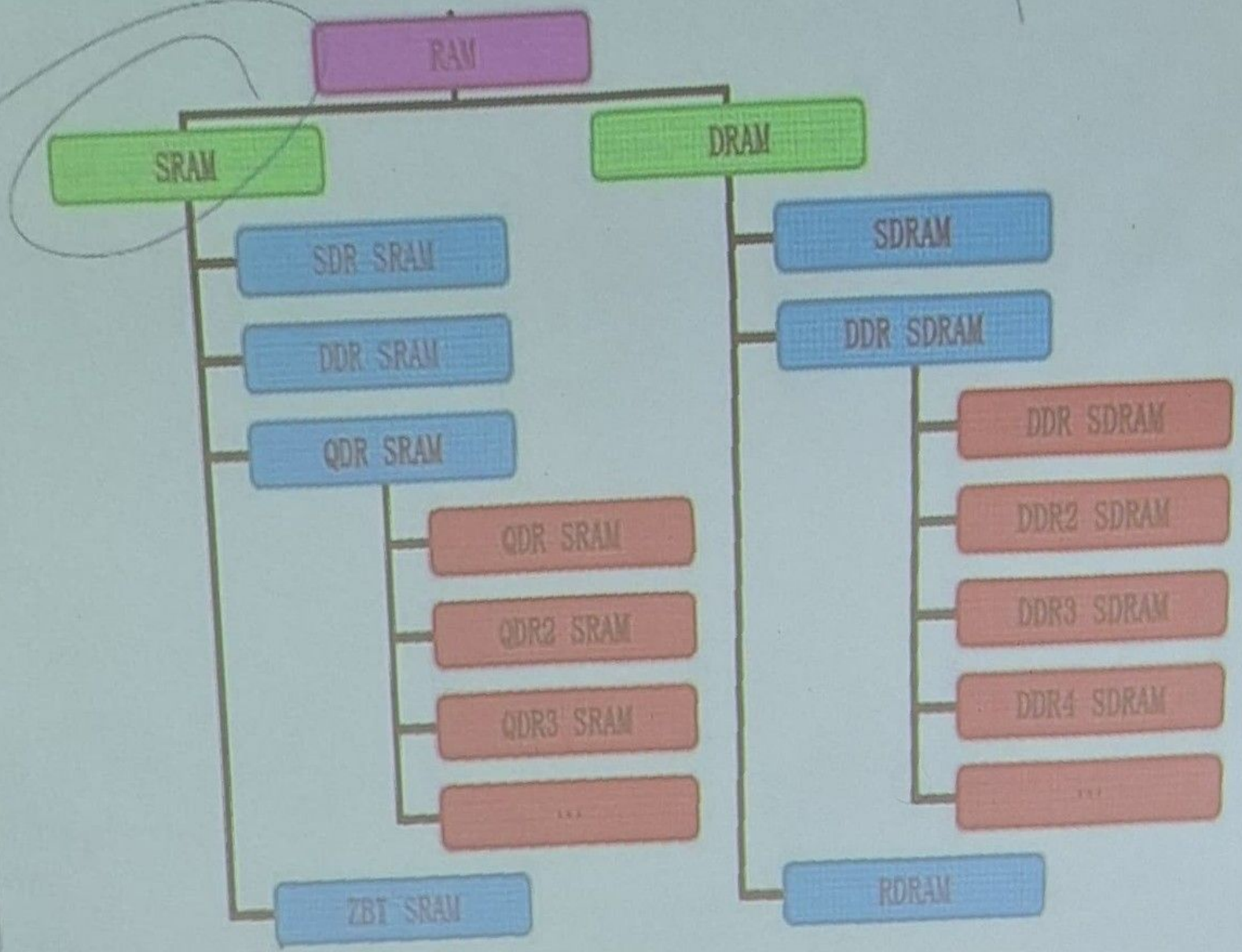
Random Access Memory

- *The most common semiconductor memory is referred to as random-access memory (RAM).*
- One distinguishing characteristic of RAM is that it is possible both to read data from the memory and to write new data into the memory easily and rapidly.
- Both the reading and writing are accomplished through the use of electrical signals.
- *Another distinguishing characteristic is that it is volatile.*
- A RAM must be provided with a constant power supply.
- If the power is interrupted, then the data are lost.
- RAM can be used only as temporary storage.

RAM Properties

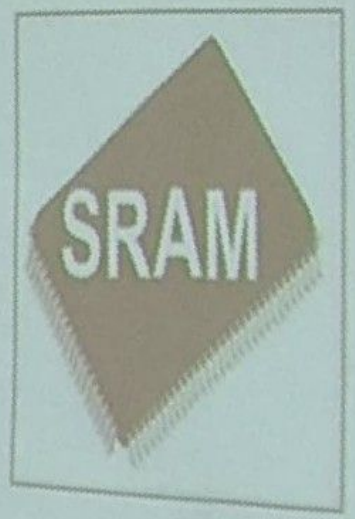
- Misnamed as all semiconductor memory is random access
- Read/Write
- Volatile
- Temporary storage
- Static or dynamic

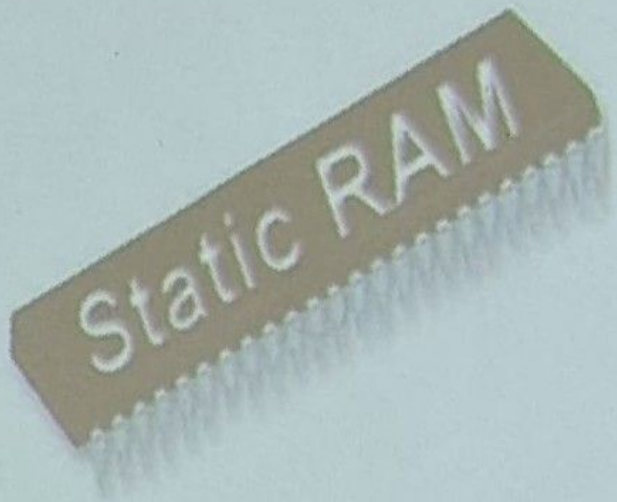
RAM Generation



SRAM

- A static RAM (SRAM) is a digital device that uses the same logic elements used in the processor.
- In a SRAM, binary values are stored using traditional flip-flop logic-gate configurations
- A static RAM will hold its data as long as power is supplied to it.



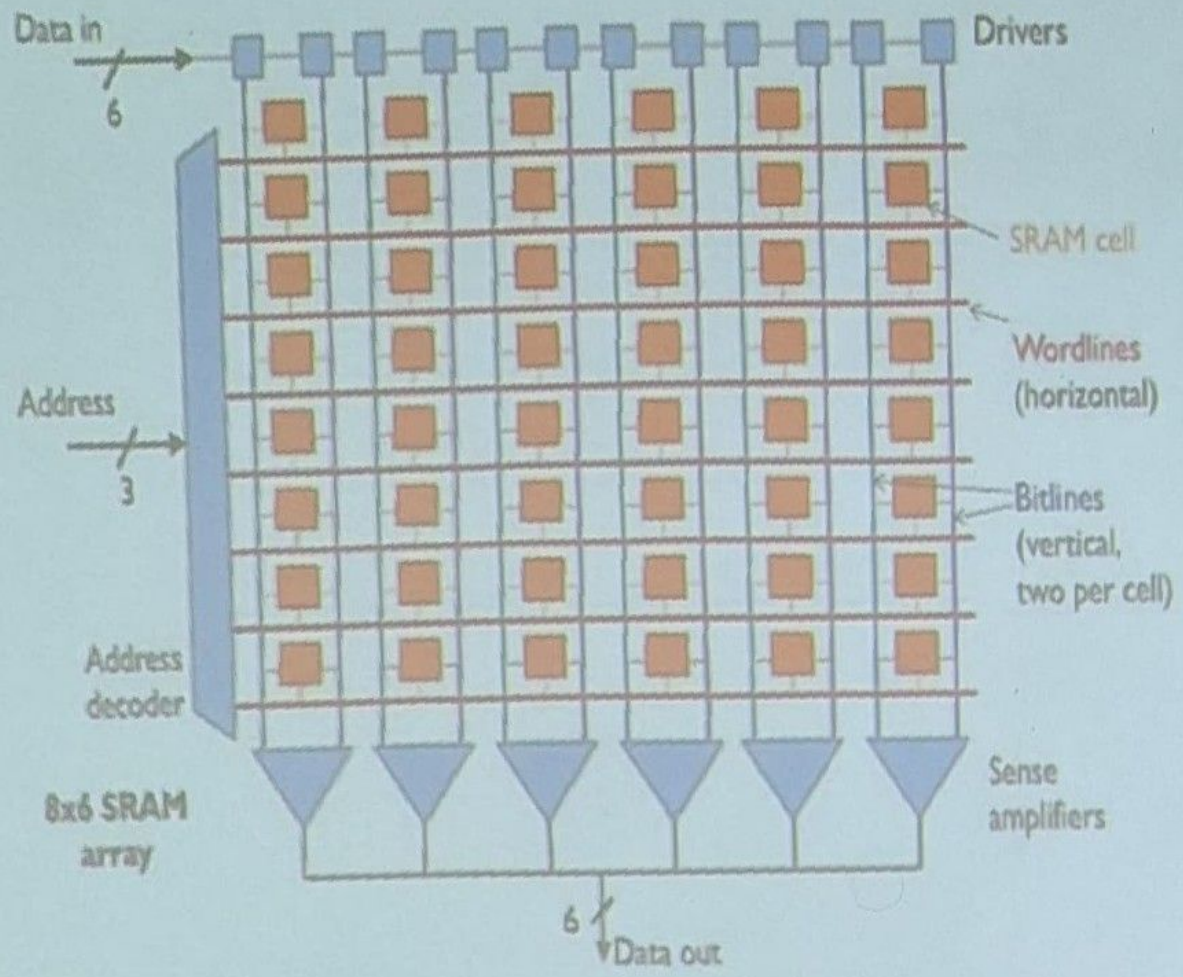


Static RAM Properties

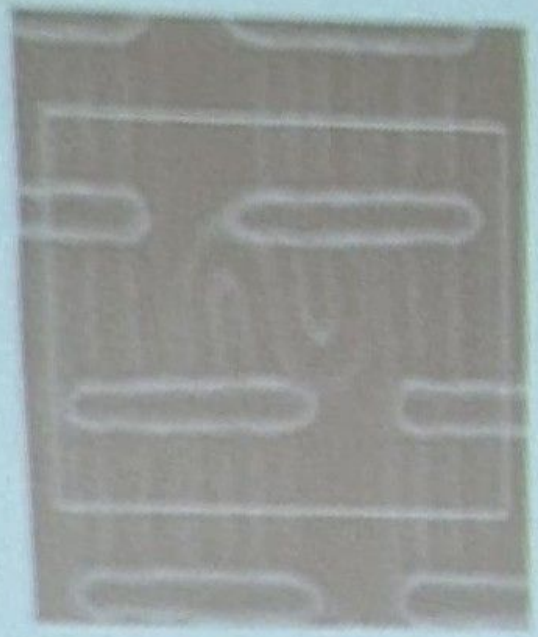
- Bits stored as on/off switches
- No charges to leak
- No refreshing needed when powered
- More complex construction
- Larger per bit
- More expensive
- Does not need refresh circuits
- Faster
- Digital
  - Uses flip-flops
- Example: Cache

SRAM Design

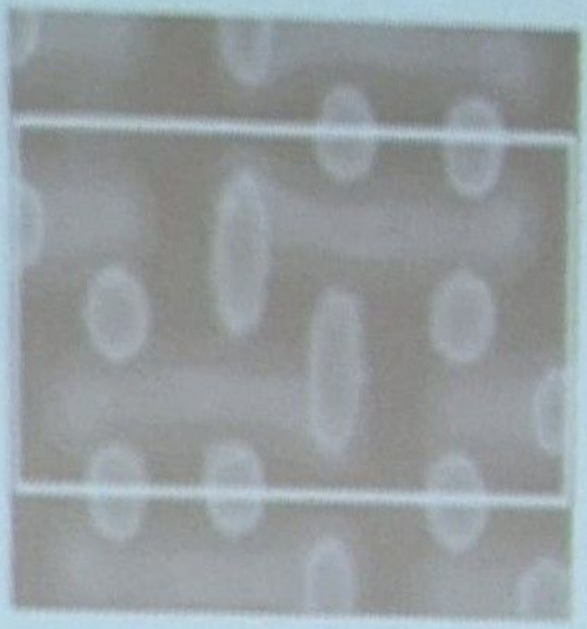
Static RAM (SRAM)



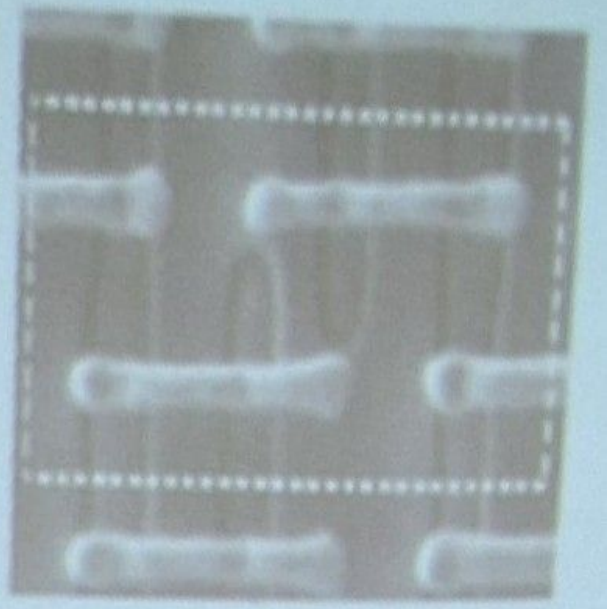
SRAM Design



TI 65nm:
0.46x1.06um²



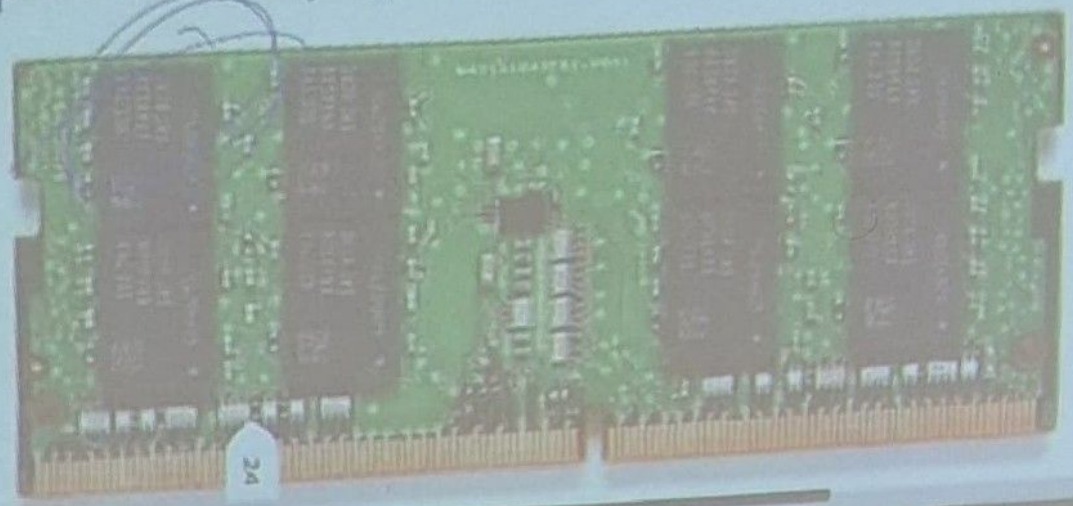
IBM 65nm:
0.41x1.25um²



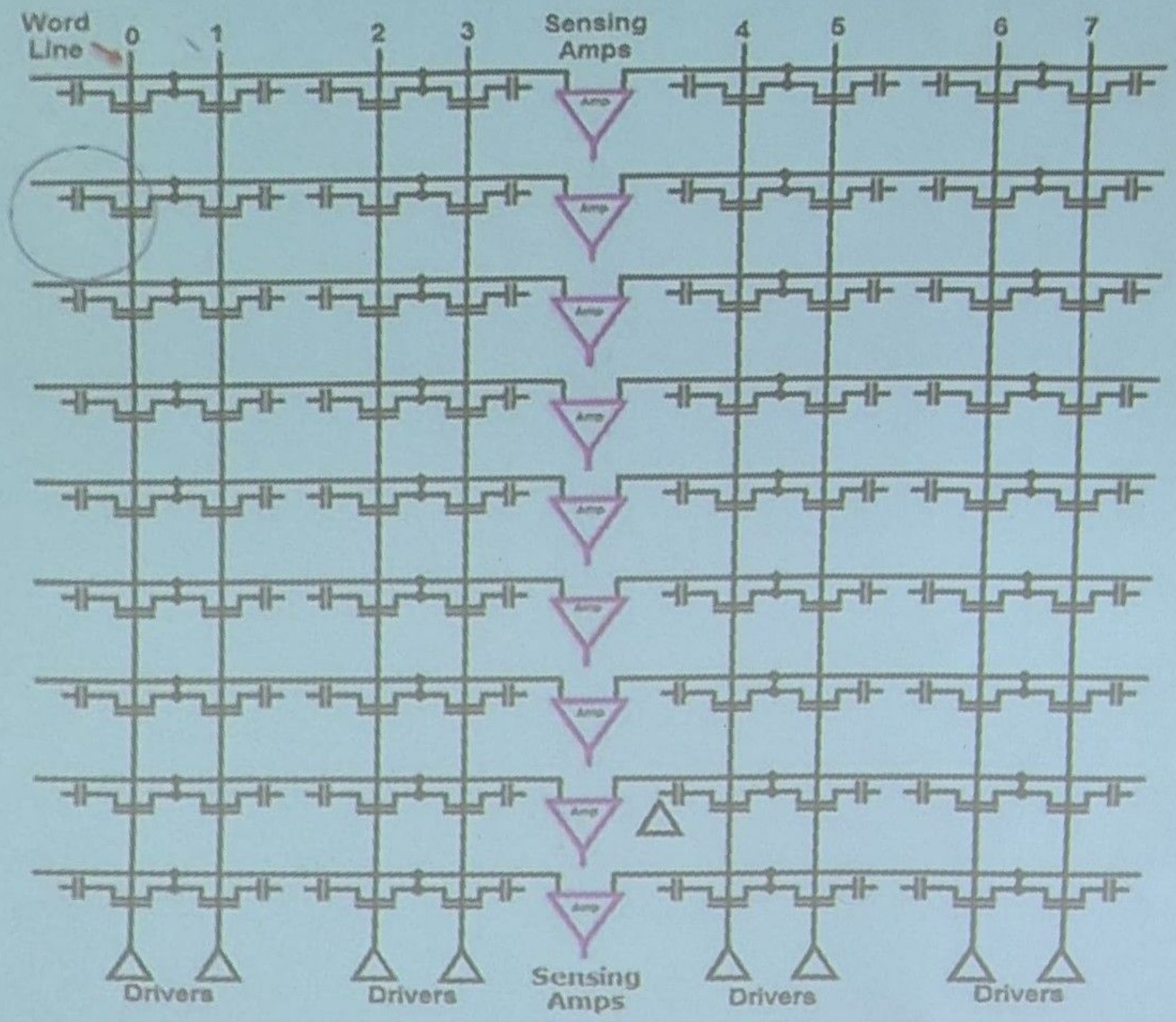
Intel 65nm:
0.46x1.24um²

DRAM

- DRAM is made with cells that store data as charge on capacitors.
- The presence or absence of charge in a capacitor is interpreted as a binary 1 or 0.
- Capacitors have a natural tendency to discharge.
- Dynamic RAMs require periodic charge refreshing to maintain data storage.
- *The term dynamic refers to this tendency of the stored charge to leak away, even with power continuously applied.*



DRAM

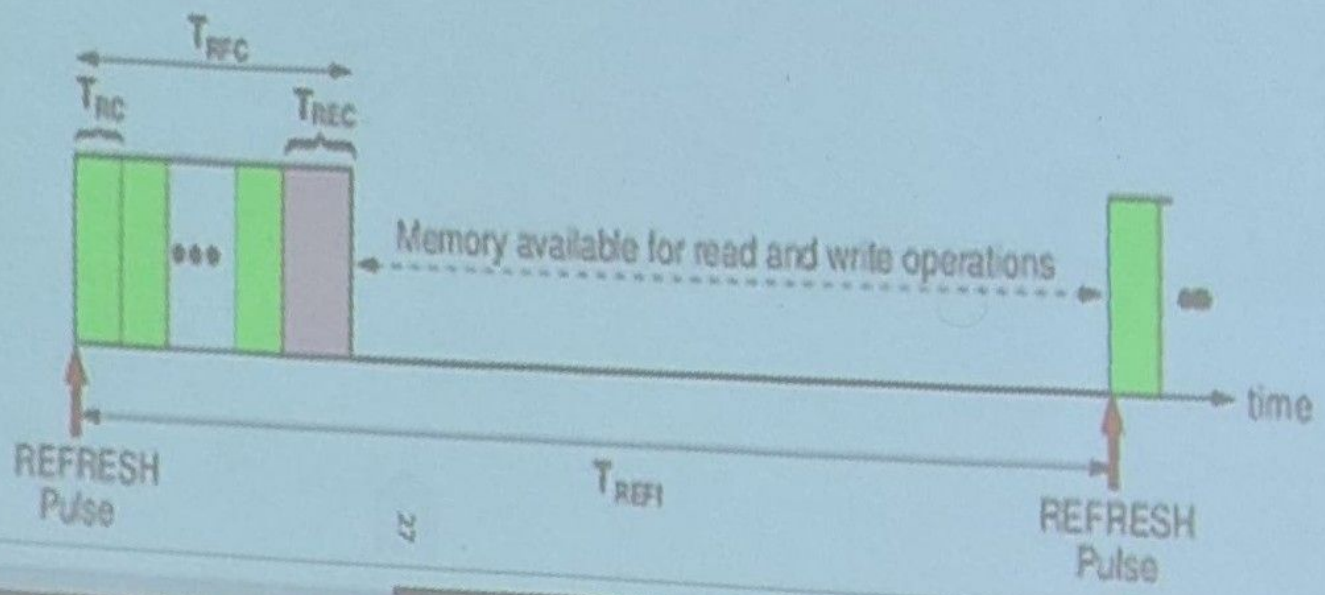

Word Line
0
1
2
3
Sensing Amps
4
5
6
7
Drivers
Drivers
Sensing Amps
Drivers
Drivers

Dynamic RAM Properties

- Bits stored as charge in capacitors
- Charges leak
- Need refreshing even when powered
- Simpler construction
- Smaller per bit
- Less expensive
- Need refresh circuits
- Slower
- Essentially analogue
  - Level of charge determines value
- **Example:** Main memory

Refreshing

- Because capacitors have a natural tendency to discharge, DRAMs require periodic charge refreshing to maintain data storage.
- Refresh circuit included on chip
- A simple technique for refreshing is "Disable chip", in effect, to disable the DRAM chip while all data cells are refreshed.
- The data are read out and written back into the same location.
- The technique takes time
- It slows down apparent performance.



Advanced DRAM Organization

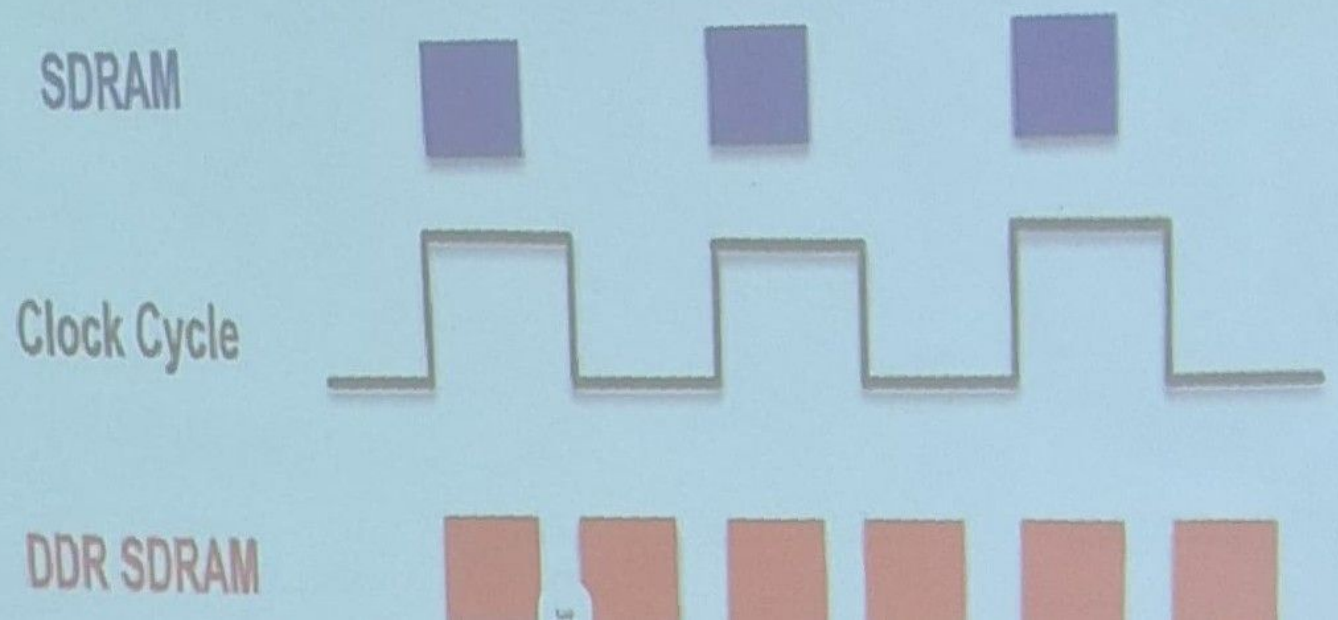
- Basic DRAM same since first RAM chips
- Enhanced DRAM
  - Contains small SRAM as well
  - SRAM holds last line read
- Cache DRAM
  - Larger SRAM component
  - Use as cache or serial buffer

DDRAM

- Computers built before 2002 generally used SDRAM.
- Memory technology took a revolutionary stride with Double Data Rate (DDR) option.
- Current computers use DDR RAM.
- Since computer hardware is all connected and interdependent, this can lead to speed increases in other components, too.
- So, upgrading computer memory is a great way to fix a slow computer.

DDRAM

- DDR, short for double data rate, was introduced in 2000 as the next generation following SDRAM.
- DDR transfers data to the processor on both the downbeat and upbeat of the clock signal, so ***twice per cycle***.
- Using both beats to transfer data makes DDR memory significantly faster than SDR memory, which uses only one edge of the clock signal to transfer data.



DDRAM

- *The process of the DDR transferring two bits of data from the memory array to the internal input/output buffer is called* **2-bit prefetch.**
- DDR transfer rates are usually between 266 and 400MT/s.
- MT/s → megatransfers (or million transfers) per second and is a more accurate measurement for the effective data rate (speed).
- Bear in mind that **double data rate** is different from **dual-channel memory.**

DDRAM Generations

- Over time, DDR technology has evolved to handle improvements to other components and has enhanced overall computer performance.
- Each generation of RAM increases speed and frequency while decreasing power consumption.
- We'll now explore and compare each generation of DDRAM.
  - DDRAM 1 (2000 – 2003)
  - DDRAM 2 (2003 – 2007)
  - DDRAM 3 (2007 – 2014)
  - DDRAM 4 (2014 – 2021)
  - DDRAM 5 (2021 – Current)

DDRAM

	SDRAM	DDR
Prefetch	1 - Bit	2 - Bit
Data Rate (MT/s)	100 - 166	266 - 400
Transfer Rate (GB/s)	0.8 - 1.3	2.1 - 3.2
Voltage (V)	3.3	2.5 - 2.6

DDRAM 2

- *DDR2 was introduced in 2003 and operates twice as fast as DDR due to an improved bus signal.*
- DDR2 uses the same internal clock speed as DDR, however, the transfer rates are faster due to the enhanced input/output bus signal.
- *DDR2 has a 4-bit prefetch, which is twice that of DDR.*
- DDR2 can also reach data rates of 533 to 800MT/s.
- *DDR2 memory can be installed in pairs to run in "dual channel mode," which can increase memory throughput even more.*

DDRAM 2

	DDR	DDR2
Prefetch	2 - Bit	4 - Bit
Data Rate (MT/s)	266 - 400	533 - 800
Transfer Rate (GB/s)	2.1 - 3.2	4.2 - 6.4
Voltage (V)	2.5 - 2.6	1.8

DDRAM 3

- *In 2007 DDR3 technology was introduced with not only 2x the bandwidth and transfer rates of DDR2, but a significant reduction in power consumption – roughly 40% compared to DDR2. .*
- This reduction from 1.8 to 1.5V meant lower operating currents and voltages, which was fantastic news for battery operated devices..
- *DDR3 transfer rates are between 800 and 1600MT/s.*
- All these improvements meant higher bandwidth and performance with less power consumption, ***making DDR3 a great memory option for laptops.***

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DDRAM 3

	DDR2	DDR3
Prefetch	4 - Bit	8 - Bit
Data Rate (MT/s)	533 - 800	1066 - 1600
Transfer Rate (GB/s)	4.2 - 6.4	8.5 - 14.9
Voltage (V)	1.8	1.35 - 1.5

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DDRAM 4

	DDR3	DDR4
Prefetch	8 - Bit	Bit per Bank
Data Rate (MT/s)	1066 - 1600	2133 - 5100
Transfer Rate (GB/s)	8.5 - 14.9	17 - 25.6
Voltage (V)	1.35 - 1.5	1.2

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DDRAM 4

- Seven years after the release of DDR3, DDR4 became available.
- The DDR4 has lower operating voltage with 1.2 V, and has higher transfer rates than previous generations, processing four data rates per cycle.
- This means DDR4 consumes less power and is faster and more efficient than DDR3.
- This newer generation also introduced bank groups to avoid having a prefetch of 16, which isn't desirable.
- With bank groups, each group can execute 8-bits of data independently from the other, so DDR4 can process multiple data requests within a clock cycle.
- DDR4 transfer rates are continually rising, as DDR4 modules can reach speeds of 5100MT/s and even higher when overclocked.
- *Crucial Ballistix MAX modules broke numerous overclocking world records in 2020.*

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DDRAM 5

- DDR5 was introduced in 2021, and is the most recent generation of memory technology, marking a revolutionary jump in architecture.
- It's arguably the biggest jump in memory technology we've seen since SDRAM.
- DDR5 brings better channel efficiency, improved power management, and optimized performance - enabling next-generation multi-core computing systems.
- DDR5 launch speeds deliver nearly double the bandwidth of DDR4.
- It also allows scaling memory performance without degrading channel efficiency at higher rates.
- These results aren't just during testing but under real-world conditions.
- DDR5 memory standard is a denser memory stick and equates to more memory capacity in your system.
- In comparison, the DDR4 stopped at 16-gigabit memory chips, but DDR5 offers up to 64-gigabit memory chips.
- Crucial DDR5 memory will operate at 4800MT/s at launch, 1.5x the maximum standard DDR4 speed.

DDRAM 5

	DDR4	DDR5
Prefetch	Bit per Bank	16 - Bit
Data Rate (MT/s)	2133 - 5100	3200 - 6400
Transfer Rate (GB/s)	17 - 25.6	38.4 - 51.2
Voltage (V)	1.2	1.1

DDR Comparisons

	DRAM	DDR	DDR2	DDR3	DDR4	DDR5
Prefetch	1-Bit	2-Bit	4-Bit	8-Bit	Bit per Bank	16-Bit
Data Rate (MT/s)	100-166	266-400	533-800	1066-1600	2133-5100	3200-6400
Transfer Rate (GB/s)	0.8-1.3	2.1-3.2	4.2-6.4	8.5-14.9	17-25.6	38.4-51.2
Voltage (V)	3.3	2.5-2.6	1.8	1.35-1.5	1.2	1.1